

HIBATULLAHI SHAKIRA

Machine Learning Engineer | AI Developer | Quantitative Finance

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PROFESSIONAL SUMMARY

Machine Learning & Software Engineer with a strong foundation in Economics and quantitative finance. Experienced in building production-grade ML pipelines and quantitative modelling systems — including Monte Carlo simulation, Real Options Theory, ensemble forecasting (Prophet + LSTM), and stochastic methods — alongside scalable backend and AI-powered software solutions. Demonstrated impact across fintech, risk modelling, financial decision intelligence, NLP/ASR, and document intelligence systems.

Genuine interest in capital modelling as a discipline, evidenced by AMSS Enterprise — a self-architected financial intelligence platform applying Real Options Theory and Prospect Theory to SME lending decisions — and by active project work in financial scenario simulation and uncertainty quantification calibrated to macroeconomic dynamics.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, SQL | Flask, Django, Spring Boot, FastAPI

ML/AI Stack: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, OpenCV, Whisper, Llama-3

Quantitative Finance: Monte Carlo Simulation, Real Options Theory, Prospect Theory, Ensemble Forecasting (Prophet + LSTM), SARIMA/ARIMA, Stochastic Modelling, Credit Risk Scoring

NLP & Speech: ASR/TTS architectures, NLTK, SpaCy, fine-tuning LLMs (LoRA), prompt engineering

MLOps & Cloud: Docker, Kubernetes, AWS (Lambda, SageMaker, EC2, S3), CI/CD (GitHub Actions), MLflow

Agent Frameworks: LangChain, LangGraph, RAG pipelines, autonomous agent architecture

Data Engineering: PostgreSQL, BigQuery, ETL pipelines, Pandas, NumPy, large-scale preprocessing

Software Engineering: REST API design, microservices, OOP, SDLC, TDD, Git, WebSockets, real-time systems

PROFESSIONAL EXPERIENCE

AI Developer

I&M Engineering Limited | Abuja, Nigeria | May 2026 – Present

- Designing and developing AI-powered document intelligence systems, including a production FastAPI service that processes German-language construction order PDFs and extracts structured data fields through custom column mapping specifications
- Diagnosed and resolved a critical production Docker container failure — identified stale image deployment (v1.1.9A) missing the application startup script, rebuilt and pushed corrected image (v1.1.10) to Docker Hub, restoring live service availability
- Performing systematic QA and output validation for AI document parsing systems, cross-referencing extracted values against master data mapping rules across 20+ line items per order document
- Collaborating with cross-functional engineering teams on AI system architecture, codebase maintenance, deployment pipeline management, and documentation per the full AI development lifecycle

Technologies: Python, FastAPI, Docker, GitHub, uvicorn, PDF processing

Software Engineer (ML-Focused)

National Centre for Artificial Intelligence & Robotics (NCAIR) | Abuja, Nigeria | Sept 2025 – Present

- Contributor to N-ATLAS v1, Nigeria's first open-source multilingual LLM (Llama-3 8B base, 400M tokens), supporting Yoruba, Hausa, Igbo, and Nigerian English — launched at UN General Assembly (UNGA80) September 2024
- Engineered ASR pipeline using fine-tuned Whisper architecture, achieving <15% WER on low-resource Nigerian languages through custom phoneme lexicon development and 50K+ hours of curated audio preprocessing
- Built production TTS inference APIs with FastAPI + Docker, reducing latency from 2.1s to 1.5s (30% improvement) and enabling real-time multilingual voice applications for government service interfaces
- Deployed speech recognition models on AWS (EC2 + S3 + Lambda) with auto-scaling to handle 10K+ daily transcription requests; model open-sourced on HuggingFace (NCAIR1/N-ATLAS)

Technologies: PyTorch, HuggingFace Transformers, Whisper, Llama-3, FastAPI, Docker, AWS (Lambda/EC2/S3)

Machine Learning Engineering Intern

Publica AI | Lagos, Nigeria | May 2025 – Sept 2025

- Architected PrepAI, an AI-powered exam prep platform for 50+ beta users, generating contextually accurate WAEC/JAMB-style questions through fine-tuned LLM and RAG pipeline integration
- Fine-tuned GPT-based model on 100K+ past exam questions + 500+ textbook chapters using HuggingFace, achieving 85% relevance score and reducing question generation time from hours to seconds
- Designed ML curriculum and mentored 45 students across 3 cohorts at CcHub Lekki AI Summer Program; curriculum adopted for CcHub East Africa (Kenya/Rwanda) — 90% completion rate
- Selected to AI Developer Panel (Nigerian Communications Commission) to evaluate 120+ candidates for national AI training programs

Technologies: HuggingFace, GPT-3.5, LangChain, Flask, PostgreSQL, Docker, RAG, NLTK, Python

Software Engineering & ML Trainee

Semicolon Africa | Lagos, Nigeria | 2023 – 2025

- Phase 1 — Software Engineering: Built full-stack applications and REST APIs using Java (Spring Boot) and Python (Flask); applied OOP, TDD, clean code principles, and Git workflow on real-world capstone projects
- Phase 2 — Data Science: Developed end-to-end data pipelines using Pandas, NumPy, and SQL; applied statistical methods and feature engineering; produced visualisations for business insights
- Phase 3 — ML Engineering: Trained and deployed ML models using Scikit-learn and PyTorch; implemented model evaluation, hyperparameter tuning, and containerised deployments with Docker
- Graduated as Best Female Student (Women in Tech Award)

Technologies: Java, Spring Boot, Python, Flask, Scikit-learn, PyTorch, PostgreSQL, Docker, Git

Software Engineer Facilitator (Part-Time)

Eyinmofe/Mofemart | Lagos, Nigeria | Aug 2024 – Aug 2025

- Designed and delivered 200+ hours of software engineering curriculum for 60+ students across 4 cohorts, teaching Java/Python, OOP, data structures, and backend fundamentals
- Mentored students from zero coding experience to building 15+ functional projects with 92% course completion rate; conducted 300+ code reviews improving debugging efficiency by ~40%

Technologies: Java (Spring Boot), Python (Flask), Git, PostgreSQL, TDD methodologies

KEY PROJECTS

AMSS Enterprise — Autonomous Financial Intelligence Platform

Lead Architect & Engineer | LangGraph, Prophet, LSTM, Monte Carlo, Django, PostgreSQL, Docker | 2026 – Present

- Designing and building an AI-native financial decision support platform for Nigerian and emerging-market SMEs — an autonomous financial intelligence system that records, predicts, and advises simultaneously
- Architected a LangGraph autonomous agent grounded in Real Options Theory and Prospect Theory with multi-step reasoning for loan decision intelligence: simulating repayment probability under macro uncertainty, computing opportunity cost, and delivering auditable plain-language recommendations
- Built ensemble cash flow forecasting layer combining Facebook Prophet and LSTM networks, calibrated for Nigerian macroeconomic dynamics — naira volatility, CBN monetary policy cycles, and informal economy patterns — achieving <12% MAPE on 6-month projections
- Implemented Monte Carlo uncertainty engine for financial scenario simulation and a Lambda Architecture separating real-time event processing from batch analytics and ML training

Technologies: LangGraph, LangChain, Facebook Prophet, LSTM (PyTorch), Monte Carlo, Django, PostgreSQL, Docker, AWS, MLflow, GitHub Actions

IPPIS Payroll Insights — Payroll Analytics Platform

Full-Stack Developer | React, Flask, PostgreSQL/SQLite, PyPDF2 | 2026

- Built a full-stack platform parsing IPPIS SoftSUITE payroll data — nominal roll spreadsheets and bulk payslip PDFs — into structured, queryable employee records, replacing manual reconciliation of payroll exports
- Developed an analytics dashboard (Recharts) surfacing salary trends, department-level payroll costs, Grade Level (GL) distributions, and top deduction categories for workforce cost analysis
- Implemented a JWT-authenticated REST API (Flask + SQLAlchemy) with an employee directory supporting search/filter by department, IPPIS number, and Grade Level, plus individual single-page payslip generation and download
- Configured a GitHub Actions CI/CD pipeline running backend tests and frontend builds automatically on every push to main

Technologies: React, Vite, Recharts, Flask, SQLAlchemy, Flask-JWT-Extended, PyPDF2, SQLite, GitHub Actions

Credit Risk Scoring System for Alternative Lending

Scikit-learn, XGBoost, Flask, PostgreSQL, Docker | 2025

- Developed credit scoring ML pipeline for SME alternative lending using logistic regression + XGBoost on 10K+ loan applications, achieving 82% classification accuracy and 0.78 AUC-ROC
- Built real-time scoring REST API (Flask) with <200ms response time, enabling instant credit decisioning through structured feature engineering (30+ financial/behavioral features)

Technologies: Scikit-learn, XGBoost, Flask, PostgreSQL, Docker

Silent Speech Interface for Accessibility

Java, WebSockets, NLP, Spring Boot | 2024

- Engineered real-time silent speech-to-text system using Java backend with WebSocket streaming, enabling accessibility applications for speech-impaired users with <500ms latency

Technologies: Java, WebSockets, NLP, Spring Boot

EDUCATION

Diploma in Software Engineering (Data Science & ML)

Semicolon Africa | 2025

- Women in Tech Award – Best Graduating Female Student
- Entrepreneurship & Business Management Certificate — Henley Business School, University of Reading

B.Sc. Economics

Lead City University | 2023

- CGPA: 4.26/5.0 (Top 5%) | Best Graduating Student – Taxation & Fiscal Policy
- Relevant coursework: Econometrics, Financial Modelling, Statistics, Calculus, Microeconomics, Derivatives

LEADERSHIP & RECOGNITION

- AI Curriculum Co-Creator & Mentor | CcHub Lekki & East Africa (Kenya/Rwanda) | 2024 – Present
- AI Developer Selection Panel | Nigerian Communications Commission Digital Innovation Park | 2024
- Featured Alumni | Semicolon Africa (3× featured) | 2024 – 2025
- Mentor | Semicolon Women in Tech Initiative | 2024 – Present